⚡ WHY LLM EVALS FAIL IN CI

Stop Burning LLM Tokens in CI/CD

One prompt tweak can silently break production AI features. Yet standard CI evaluation playbooks blow through API quotas and stall pull requests.

Here is the **Two-Tier Quality Gate** architecture that runs on free-tier limits and catches regressions before merge.

🎯 **Who This Is For**

Engineering Leaders, Tech Leads, and AI Platform Teams shipping LLM features without wanting a runaway CI bill.

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WHAT STANDARD PLAYBOOKS RECOMMEND

Heavy Frameworks, Retries & Single Keys

Most CI/CD guides for LLM apps assume unlimited paid API budgets and treat every test case as an expensive cloud call.

🔁 **Multi-Pass Voting (`repeat: 3`)**

To handle non-deterministic outputs, teams run every test 3–5 times in parallel—instantly tripling API calls and rate-limit pressure.

🔑 **Single-Provider Bottlenecks**

Both the app-under-test and the LLM judge hit the exact same API key and quota pool on every pull request.

📦 **Framework Bloat**

Heavyweight eval wrappers pull in hundreds of MBs of dependencies just to assert basic schema and policy rules.

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3 Ways Standard CI Evals Break at Scale

Even a tiny 15-case regression suite (1 generation + 1 judge call = 30 requests) exposes three structural flaws:

🛑 **1. Instant Rate-Limit Deadlocks (HTTP 429)**

Free-tier preview models (like Gemini Flash at **20 requests/day**) halt on the very first test case when generation and judging share one quota.

💥 **2. Silent JSON Schema Drift**

Asking an LLM to *"Output valid JSON"* still drops required fields in production—crashing downstream microservices that expect strict contracts.

🔒 **3. Broken External Contributor PRs**

GitHub Actions strips API secrets on fork PRs. Without a zero-token local gate, external contributions fail CI before review.

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ARCHITECTURAL SHIFT

Decouple Structural From Semantic Gates

Never burn LLM tokens on what deterministic code can reject in 10 seconds—and never share a quota pool between your app and your judge.

1️⃣ **Tier 1: Zero-Token Fast Gate (< 10s)**

Validates JSON schemas, scans for leaked PII (cards, SSNs, keys), and enforces token budgets locally. Runs on 100% of PRs—even forks.

2️⃣ **Tier 2: Dual-Provider Semantic Judge**

Routes app inference to a high-quota provider (Groq: 14,400 req/day) and isolates quality judging on a separate provider (Gemini).

💾 **Deterministic SHA-256 Cache Replay**

Unchanged prompts replay from CI cache in under 1 second—consuming **zero API tokens** across repeated commits.

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Grammar-Constrained Native SDK Judge

Replace fragile JSON prompting and heavyweight wrappers with schema-enforced native SDK calls.

❌ BEFORE: SINGLE QUOTA + FRAGILE JSON PROMPT

```
# Both app & judge hit the same 20 req/day limit
app_model = "gemini-3.8-flash"
judge     = "gemini-3.8-flash"  # 429 crash + missing fields
```

✅ AFTER: DECOUPLED PROVIDERS + ENFORCED SCHEMA

```
# 1. Inference on Groq (14,400 RPD) + SHA-256 cache
# 2. Judge on Gemini with strict Pydantic grammar:
config = {
    "response_mime_type": "application/json",
    "response_schema": TriageResult, # 100% field guarantee
    "temperature": 0.0,
}
```

⚡ **70 Lines vs. Hundreds of MBs**

Using the native provider SDK directly eliminates third-party eval framework bloat and gives bit-exact schema validation.

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15-Ticket Suite: Before vs. After

Tested across real enterprise scenarios: P0 outages, PII leaks, billing disputes, and prompt injections.

CI CONFIGURATION	RUNTIME	RESULT
Single-Provider Shared free-tier quota	Halted	Crashed on Case 1 (429 daily limit hit)
Two-Tier Cold Run Groq + Gemini Judge	3m 08s	**100%** schema pass; caught **6** subtle prompt regressions
Two-Tier Cached CI GitHub Actions cache	< 1s	**0** API tokens consumed on unchanged prompts

🎯 **High-Signal Catch Rate**

Caught silent escalation drift (e.g., treating routine phishing questions as P0 emergencies) before a single user was impacted.

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3 Rules to Audit In Today's AI PRs

Treat prompts as production software without blowing your CI budget:

- ✓ **Gate with Tier 1 first (<10s, 0 tokens):** Enforce schemas, PII scans, and token limits before calling any LLM.
- ✓ **Enforce `response_schema`:** Use grammar-constrained decoding instead of fragile JSON prompting.
- ✓ **Decouple & Cache:** Separate inference and judge providers, persist SHA-256 caches, and skip live keys on fork PRs.

Read the Full Technical Paper

Inspect the complete Two-Tier CI/CD architecture and 6-case regression breakdown at **avilano.dev** (link in first comment).

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